

Homework 1:

Perturbative Renormalization of QED at one loop

- a) Write down the bare action for QED in $3+1$ dimensions
- b) Write it in terms of a renormalized action & counter terms
- c) Derive the Feynman rules for Renormalization of QED at 1-loop
- d) Compute the TPI diagram for vacuum polarization in dimensional regularization at 1-loop
- e) Compute the TPI diagram for fermion propagator at 1-loop

f) Compute the 1PI diagram for vertex interaction at one-loop.

g) Use the answers to fix the counter terms.